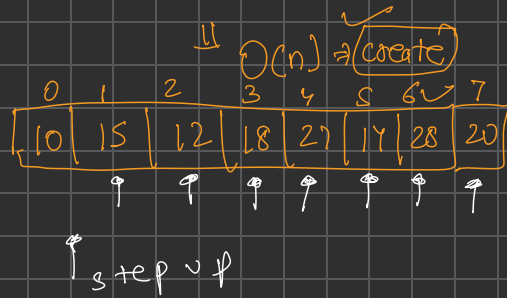
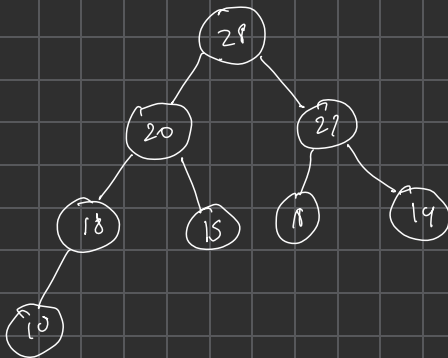
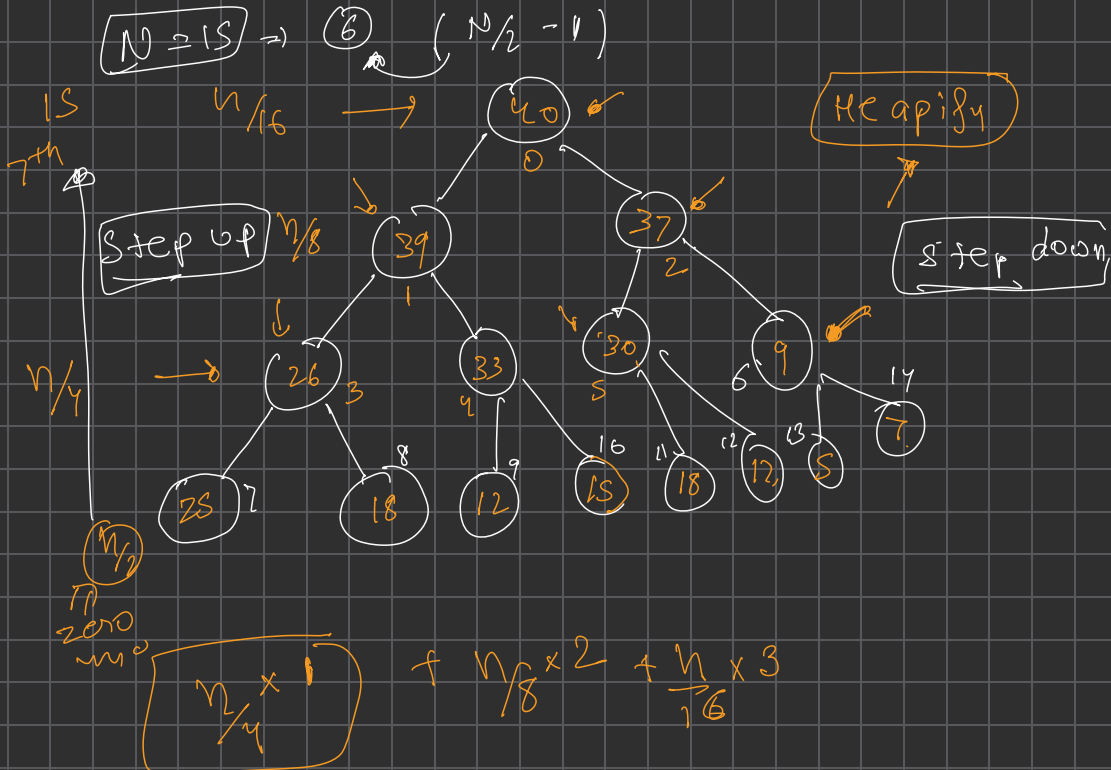




Heap: Max Heap: $\Rightarrow \Theta(n) \Rightarrow n \log n$



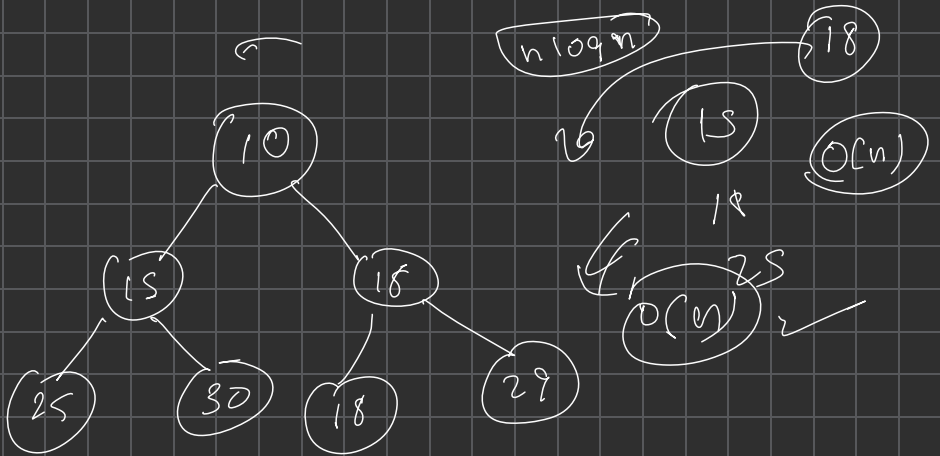


$$\frac{n}{4} \times 1 + \frac{n}{8} \times 2 + \frac{n}{16} \times 3 + \frac{n}{32} \times 4 + \dots 1 \times \log n$$

$$n \left(\frac{1}{4} + \frac{2}{8} + \frac{3}{16} + \frac{4}{32} + \frac{5}{64} + \dots 1 \right)$$

$$O(n) \rightarrow 0.25 + 0.25 + 0.18 + \dots \rightarrow$$

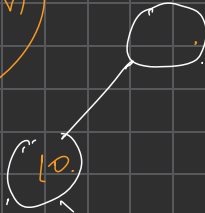
for $i = n/2 - 1; i \geq 0; i--$
 {
 heapify(arr, i, n);
 }



$n \log n \times n$

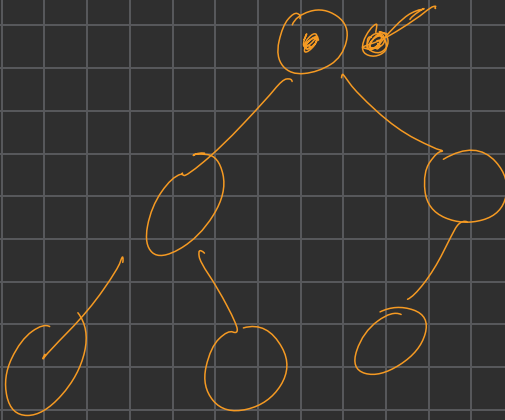
$n \log n$

Heap sort $\Rightarrow$ Max element



n elements
 $\downarrow$
 Max Heap
 $O(n)$

60 40 30 10 17 12 13

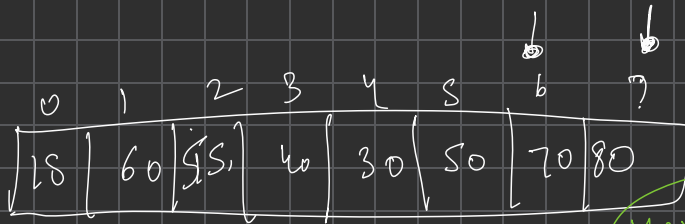
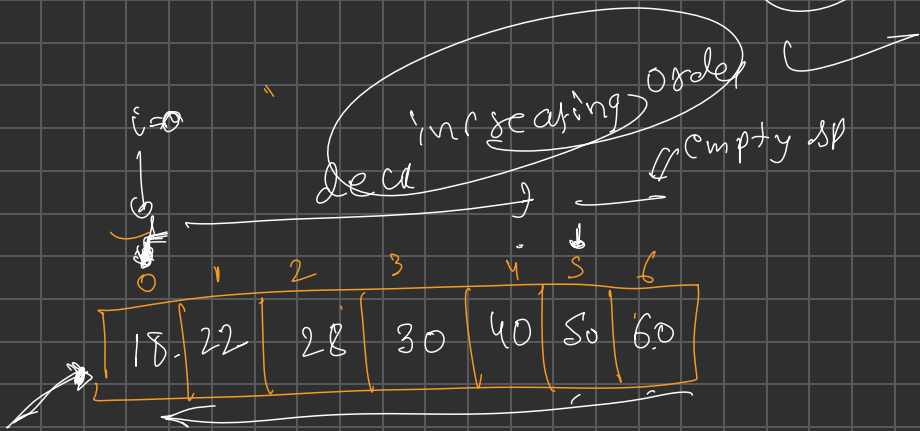


$n \log n$

30

18

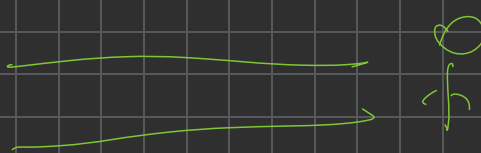
60
50
↓
lag
array



priority - queue

Max Heap

↓
900



$$\log_2 7 \Rightarrow$$

$$\frac{7}{2} \Rightarrow 3 \frac{1}{2}$$

$$1+1$$

$$2$$

$$16$$

$$\log_2 N$$

$$\log_2 8 \Rightarrow 3$$

$$\log_2 9 \Rightarrow 3.01$$

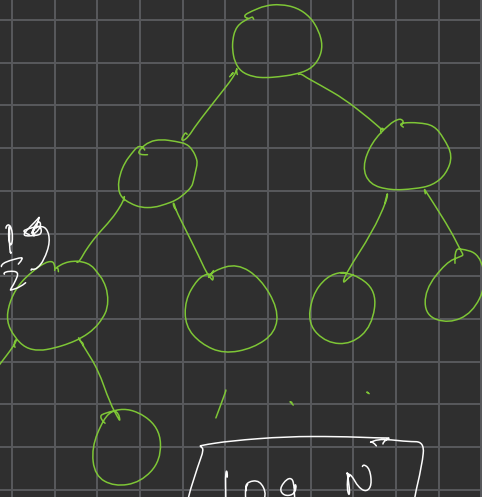
direct

$$\log_2 16 \Rightarrow 4$$

$$\log_2 12 \Rightarrow 4.01$$

$$2^x \Rightarrow 7$$

$$2^{x-1} \Rightarrow 3$$



180°	2° ✓
0	0
2	1
3	1
4	2
5	2
6	2
7	2
8	3
9	3
10	3
16	4

$$\frac{16}{2} = 8 \Rightarrow \frac{8}{2} = 4 \Rightarrow \frac{4}{2} = 2 \Rightarrow \frac{2}{2} = 1$$

$$1 + 1 + 1 + 1 = 4$$

$$[\dots 28]$$

Minimum
cost

$$\text{Cost} = 6 + 12 + 16 + 28 \Rightarrow 62$$

↑
min

↑ day

$\log n + \log n$
 $n \log n \rightarrow \text{Complexity}$
 Cost = 0
 $6 + 12 + 16 + 28$
 $2 + 9 = 11$
 $12 + 16 + 28$
 length 28
 stop type
 Min heap
 28 length

$k \log n$
 $[5, 8, 9, 4, 3]$, $k = 4$
 $\log + n \log n$
 maximum no of
 $100 + 64 + 25 + 10$
 199
 29
 36
 20
 1
 0
 199